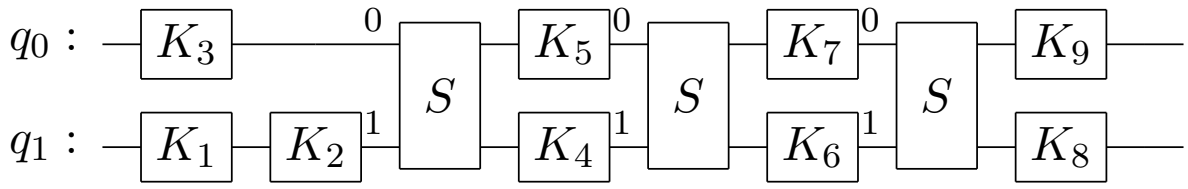
Exact symbolic SQiSW circuits: completing the outer local frames

Symbolic decomposition construction ledger

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The new constructor implements every real member of the five families below with three SQiSW pulses, in each of the two supplied bases, under both phase policies. The previous scored constructions used six. The circuits include explicit symbolic outer frames and all degeneracy and periodic-angle branches. They do not call a numerical KAK decomposition.

Target family	Previous pulses	New pulses	New local gates
Canonical	6	3	9 exact / 8 projective
fSim	6	3	9 exact / 8 projective
PhasedFSim	6	3	9 exact / 8 projective
SWAPPow	6	3	9 exact / 8 projective
TK2	6	3	9 exact / 8 projective



Qiskit drawing of the actual phase-exact Canonical construction: S is native SQiSW and K_j are the nine derived U_3 gates, in archived operation order. The formulas below define the local matrices; the scalar program records their exact Euler angles. The projective circuit has eight local gates and a virtual scalar in its metadata.

Each row applies to both `sqrt_iswap_u` and `google_sqrt_iswap`, giving twenty phase modes. Three is the *minimum uniform family budget*: every family has a specialization in the SWAP class, which cannot use two SQiSW pulses [1]. This is not a claim that every specialization needs three pulses. Identity, native-pulse, and two-pulse specializations can require fewer.

The parameter-sensitive local score is explicitly *not evaluated* for these shared-expression circuits. Assigning costs to temporary symbol names would incorrectly score aliases rather than their defining expressions. The existing scored archive is preserved; the separate construction-frontier archive supplies the new pulse bounds and exact replay certificates. No new global local-score or literature-novelty claim is made here.

1 Conventions and the two-pulse middle chart

Write

$$\mathcal{A}(x,y,z)=\exp(i(xXX+yYY+zZZ)),\qquad S=\mathrm{SQiSW}=\mathcal{A}(\pi/8,\pi/8,0).$$

First assume $\pi/4\ge x\ge y\ge|z|$ and $x\ge y+|z|$. The middle chart of [1] can be expressed without inverse cosines. Put

$$\begin{aligned}
&X=\cos 2x, && Y=\cos 2y, && Z=\cos 2z,\\
&d=X-Y+Z, && r=\sqrt{X^2+Y^2+Z^2-2XYZ-1}, &&\\
&C=(1+d-r)/2, && V=(1-d-r)/2, && D=(1+X)(1+Z)(1-Y)/2.
\end{aligned}$$

On the regular chart define

$$c=\sqrt{C},\quad t=\sqrt{1-C},\quad v=\sqrt{V},\quad u=\mathrm{sgn}_+(z)\sqrt{D}/c,\quad w=\sqrt{XYZ}/c,$$

where $\mathrm{sgn}_+(0)=1$. Then

$$c^2+t^2=1,\quad u^2+v^2+w^2=1,\quad U=S\big((uI+ivX+iwZ)\otimes(cI+itX)\big)S.$$

The only $c=0$ point in this chamber is the CNOT corner $(x,y,z)=(\pi/4,0,0)$. There the explicit branch is $(c,t,u,v,w)=(0,1,0,0,1)$; no division by zero is evaluated.

The domain inequalities follow from $0\le X\le Y,Z\le 1$ and $x\ge y+|z|$. In particular

$$r^2=(X-YZ)^2-(1-Y^2)(1-Z^2)\ge 0,$$

and

$$\begin{aligned}
(1-d)^2-r^2&=2(1-X)(1+Y)(1-Z)\ge 0,\\
(1+d)^2-r^2&=4(D+XYZ)\ge 0.
\end{aligned}$$

Thus every square root denotes its real nonnegative branch. The identities

$$\begin{aligned}
C(1-V)&=D+XYZ,\\
4(1-C)VD&=(1-X^2)(1-Y^2)(1-Z^2)
\end{aligned}$$

give the required normalization and the sign of the spin trace.

2 Exact local-equivalence certificate

For $W\in SU(4)$ let

$$\Gamma(W)=W(YY)W^T(YY).$$

The implemented polynomial checks establish

$$\begin{aligned}
\tfrac{1}{4}\operatorname{tr}\Gamma(U)&=c^2w^2+2ictuv=XYZ+i\sin 2x\sin 2y\sin 2z,\\
e_2(\Gamma(U))&=4c^4-8c^2u^2+4v^4-8v^2+2\\
&=2(\cos 4x+\cos 4y+\cos 4z).
\end{aligned}$$

Here $e_2(M)=((\operatorname{tr}M)^2-\operatorname{tr}(M^2))/2$. Both spin matrices are unitary with determinant one, so the third characteristic coefficient is the conjugate of the trace and the fourth is one. Their full characteristic polynomials therefore agree. The standard two-qubit local-equivalence theorem [2] supplies local $SU(2)$ factors; the next section recovers those factors explicitly. No numerical comparison is used as this certificate.

3 Recovering the local factors through a real 3×3 tensor

The spin matrix has the Pauli expansion

$$\Gamma(U) = k_0 I + \sum_{i,j=1}^{3} T_{ij} \sigma_i \otimes \sigma_j.$$

There are no one-body coefficients. For the target, the tensor is diagonal, with

$$k_i = \cos 2x_i \sin 2x_j \sin 2x_k + i \sin 2x_i \cos 2x_j \cos 2x_k, \qquad \{i,j,k\} = \{1,2,3\}.$$

Under local conjugation the tensor transforms by real rotations on its two indices. Hence $T = W \operatorname{diag}(k_i) V^T$ for $W, V \in SO(3)$. Write $a_i = \Re k_i$, $b_i = \Im k_i$. The product $a_i b_i = q$ is independent of i. Consequently

$$H = (\Re T)^T \Re T - (\Im T)^T \Im T = V \operatorname{diag}(h_i) V^T, \qquad h_i = a_i^2 - b_i^2.$$

These are known symbolic eigenvalues; a polynomial root solver is unnecessary. For each label i construct its spectral projector as

$$P_i = \prod_{j \neq i,\ h_j \neq h_i} \frac{H - h_j I}{h_i - h_j}.$$

Repeated nonmatching eigenvalues can repeat a projector factor harmlessly. The implementation tests every eigenvalue-multiplicity pattern exactly. A column $P_i e_p / \sqrt{(P_i)_{pp}}$, choosing the largest diagonal, gives a unit vector. If labels zero and one share an eigenspace, subtract $v_0 v_0^T$ before choosing v_1. Set $v_2 = v_0 \times v_1$.

For nonzero k_i, the matching other column is

$$w_i = \frac{a_i \Re T + b_i \Im T}{a_i^2 + b_i^2} \, v_i.$$

If a coefficient vanishes, complete its frame with cross products and a projected coordinate vector. All eight zero/nonzero patterns are explicit. A collision $h_i = h_j$ implies $k_i^2 = k_j^2$, since $\Im(k_i^2) = 2q$. Thus $k_i = \pm k_j$, and the degenerate eigenspace presents no missing branch. The local-equivalence certificate fixes the orientation in the full-rank case; cross products fix it in deficient-rank cases.

Lift each real rotation to a unit quaternion, taking the largest of

$$q_0^2 = (1 + \operatorname{tr} R)/4, \qquad q_j^2 = (1 + 2R_{jj} - \operatorname{tr} R)/4.$$

That pivot is at least $1/4$. Antisymmetric and symmetric matrix entries recover the other quaternion components. The implementation verifies these identities against the quaternion Pauli action as polynomial equalities. The corresponding $SU(2)$ matrix is $Q = q_0 I - i \sum_j q_j \sigma_j$.

Let $L = Q(W) \otimes Q(V)$ and set

$$R = \mathcal{A}^\dagger L^\dagger U.$$

Then $\Gamma(R) = I$ and $\det R = 1$. In the magic basis this is a proper real orthogonal matrix, so the $SU(2) \times SU(2)$ double cover of $SO(4)$ makes R local. Its coefficients in the orthonormal quaternion basis $(I, -iX, -iY, -iZ)^{\otimes 2}$ form a real rank-one outer product. A normalized nonzero column recovers the first quaternion and contraction with that column recovers the second. Thus the final, literal identity is

$$\boxed{\mathcal{A} = L^\dagger S (C_0 \otimes C_1) S R^\dagger.}$$

4 Three pulses and arbitrary real angles

For arbitrary coordinates, fold each angle into $[-\pi/4, \pi/4)$ using

$$n_i = \left\lfloor 2x_i/\pi + \tfrac{1}{2} \right\rfloor, \qquad e^{i\pi n_i \sigma_i \sigma_i/2} = e^{-i\pi n_i/2}(R_i(-\pi n_i) \otimes R_i(-\pi n_i)).$$

Simultaneous local Clifford permutations order their absolute values; a Pauli frame on one wire makes the first two signs positive. The scalar $e^{-i\pi \sum_i n_i/2}$ is retained explicitly.

For canonical coordinates choose a commuting one-pulse kernel

$$b = \begin{cases} (0, \pi/8, -\pi/8), & y > \pi/8,\ z < 0, \\ (0, \pi/8, \pi/8), & y > \pi/8,\ z \geq 0, \\ (-\pi/8, \pi/8, 0), & y \leq \pi/8. \end{cases}$$

Each is a fixed local conjugate of S. Canonicalizing the residual $(x, y, z) - b$ puts it in the two-pulse chamber. To verify the inequality, measure $x, y, |z|$ in units $\pi/8$ as a, b, c with $0 \leq c \leq b \leq a \leq 2$. The four residual differences between its largest absolute coordinate and the sum of the other two are

$$a + (b - c), \quad (2 - a) + (b - c), \quad (a - b) + c, \quad (a - b) + (2 - c),$$

on the branches $(b \leq 1, a < 1)$, $(b \leq 1, a \geq 1)$, $(b > 1, c \leq 1)$, $(b > 1, c \geq 1)$, respectively. Each is nonnegative. Multiplying the one-pulse kernel and the two-pulse residual yields three pulses and four local layers, after fusing adjacent local factors.

Each unit quaternion has an explicit polar U_3 chart, including zero diagonal and off-diagonal radii. Projective output records its scalar as virtual metadata. Exact output absorbs that scalar into a two-U_3 local word, replacing one existing local gate. The PhasedXZ conversion has its own checked scalar. No physical GP operation appears in either basis.

5 Reproduction and limitations

The arithmetic graph is an exact representation with shared subexpressions; its bindings introduce no independent parameters. Lazy branches avoid evaluating unused singular denominators. The implementation files are `symbolic_sqrt_iswap_cartan.py` and `symbolic_expression_program.py`. Generate and replay the archive with

```
python examples/decomposition/symbolic_sqrt_iswap_frontier.py \
  --output benchmarks/symbolic/sqrt_iswap_frontier_20261107.json.gz
python examples/decomposition/symbolic_sqrt_iswap_frontier.py \
  --verify benchmarks/symbolic/sqrt_iswap_frontier_20261107.json.gz
```

The archive records source hashes, the code commit, twenty regenerated certificates, and matching uniform-budget witnesses. The independent regressions include full matrices at poles, repeated eigenvalues, fold boundaries, negative and out-of-chamber angles, both native bases, both phase policies, and all five free families. Regression samples supplement the component identities and local-equivalence theorem; they do not certify a symbolic family.

The remaining work is substantial local-score optimization and broader global-optimality proofs. Neither the three-pulse construction nor the earlier all-phase Clifford proof establishes global optimality throughout the whole 327×32 grid. The existing scored archive and atlas retain their previous version until shared-expression scores are evaluated under the existing objective without alias substitution.

References

[1] C. Huang et al., *Towards ultra-high fidelity quantum operations: SQiSW gate as a native two-qubit gate*, arXiv:2105.06074.

[2] Y. Makhlin, *Nonlocal properties of two-qubit gates and mixed states, and the optimization of quantum computations*, arXiv:quant-ph/0002045.